Calculus and Linear Algebra

Inflection Points and Complete Function Studies

Lecture 24

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Today's plan

First hour

- What an inflection point really is.
- Horizontal, oblique and vertical tangents.
- Ascending and descending inflections.
- Reliable tests and common mistakes.

Second hour

Two complete function studies:

$$f_1(x) = \exp\left(-\frac{x}{(x-2)^2}\right),$$

$$f_2(x) = \sqrt{|x-2|} - |x|.$$

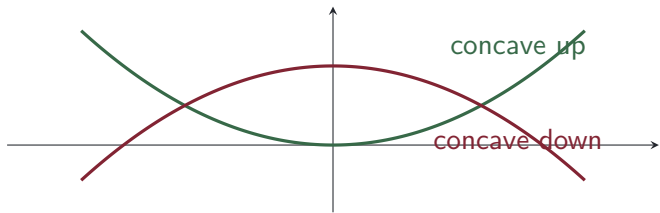
The goal is to close the qualitative study of one-variable functions with a precise treatment of inflection points.

Recall: concavity

Assume that f is twice differentiable on an interval.

$$f''(x) > 0 \implies \text{concave up / convex}$$

$$f''(x) < 0 \implies \text{concave down}$$



Inflection point: the essential definition

Definition

A point $P = (x_0, f(x_0))$ is an **inflection point** when the graph changes concavity as it passes through x_0.

Therefore the key question is not simply

$$f''(x_0) = 0?$$

but rather

Does the concavity change across x_0?

- x_0 must belong to the domain of f.
- $f''(x_0)$ may be zero, or may even fail to exist.
- The sign of f'' on the two sides is usually the safest test.

$f''(x_0) = 0$ is not enough

Compare

$$f(x) = x^3, \qquad g(x) = x^4.$$

At $x = 0$ both satisfy

$$f''(0) = g''(0) = 0.$$

But

$$f''(x) = 6x \quad \text{changes sign,}$$

whereas

$$g''(x) = 12x^2 \geq 0 \quad \text{does not change sign.}$$

Conclusion

$(0, 0)$ is an inflection point of x^3, but not of x^4.

Where should we look for inflection points?

Candidates typically occur where

$$\boxed{f''(x) = 0}$$

or where f'' does not exist but f remains continuous. For every candidate x_0:

1. verify that $x_0 \in D_f$;
2. study the sign of f'' immediately to the left and right;
3. only if the sign changes, declare an inflection point;
4. then classify the tangent and the direction of the inflection.

Two independent classifications

An inflection point has **two different attributes**.

Tangent type	**Change of concavity**
Horizontal: $f'(x_0) = 0$	Ascending: concave down $\to$ concave up
Oblique: $0 < \lvert f'(x_0)\rvert < \infty$	Descending: concave up $\to$ concave down
Vertical: vertical tangent at x_0	

Important

"Ascending/descending" describes the *change of concavity*, not whether the function itself is increasing or decreasing.

Ascending versus descending

Ascending inflection

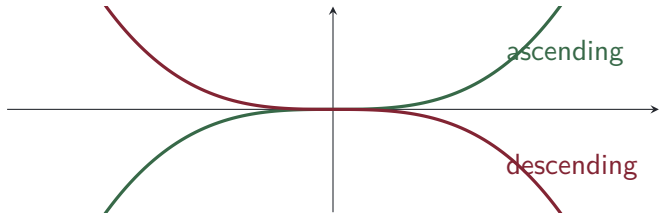
$$f'': \quad - \longrightarrow +$$

Concave down on the left, concave up on the right.

Descending inflection

$$f'': \quad + \longrightarrow -$$

Concave up on the left, concave down on the right.



Horizontal ascending inflection

Consider

$$f(x) = x^3.$$

Then

$$f'(x) = 3x^2, \qquad f'(0) = 0,$$

so the tangent at the origin is horizontal. Moreover

$$f''(x) = 6x : \qquad - \longrightarrow +.$$

Therefore

$$\boxed{(0,0) \text{ is a horizontal ascending inflection point.}}$$

Notice that f is increasing on both sides: "ascending" here refers to concavity, not monotonicity.

Horizontal descending inflection

For

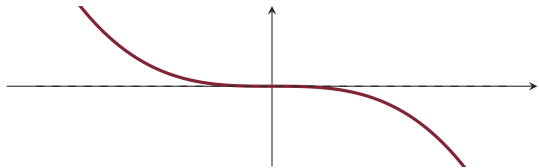
$$f(x) = -x^3,$$

we have

$$f'(0) = 0, \qquad f''(x) = -6x : \quad + \longrightarrow -.$$

Hence

$(0, 0)$ is a horizontal descending inflection point.



Oblique inflection points

An inflection point is **oblique** when the tangent has finite non-zero slope. Example:

$$f(x) = x^3 + x.$$

At $x = 0$:

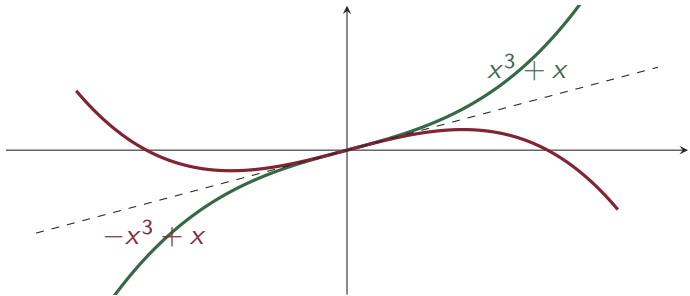
$$f'(0) = 1 \neq 0, \qquad f''(x) = 6x : \ - \to +.$$

Thus the origin is an **oblique ascending** inflection point, with tangent

$$\boxed{y = x.}$$

Changing x^3 into $-x^3$ produces an oblique descending inflection.

Oblique ascending vs descending



Both functions have tangent $y = x$ at the origin, but the change of concavity is opposite.

Vertical inflection points

An inflection point may occur even when the derivative is not finite. Consider

$$f(x) = \sqrt[3]{x}.$$

For $x \neq 0$,

$$f'(x) = \frac{1}{3x^{2/3}} > 0, \qquad f'(x) \to +\infty \quad (x \to 0^{\pm}).$$

Thus the tangent at the origin is vertical. Moreover

$$f''(x) = -\frac{2}{9x^{5/3}}$$

is positive for $x < 0$ and negative for $x > 0$. Therefore $(0, 0)$ is a **vertical descending inflection point**.

Vertical ascending inflection

For

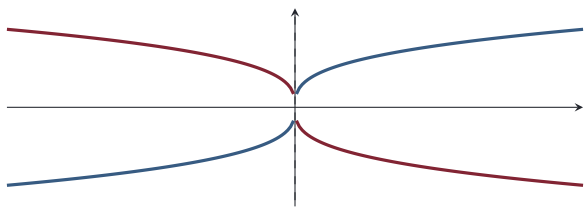
$$g(x) = -\sqrt[3]{x},$$

the tangent at the origin is again vertical, but now

$$g''(x): \quad - \longrightarrow +.$$

Hence

$(0,0)$ is a vertical ascending inflection point.



A useful third-derivative test

Assume f is sufficiently smooth around x_0 and

$$f''(x_0) = 0, \qquad f'''(x_0) \neq 0.$$

Then f'' changes sign at x_0, so x_0 is an inflection abscissa. Moreover:

$$\boxed{f'''(x_0) > 0 \implies \text{ascending inflection}}$$

$$\boxed{f'''(x_0) < 0 \implies \text{descending inflection}}$$

The tangent type is still determined separately by $f'(x_0)$.

Classification table

	Ascending	**Descending**
Horizontal	$f' = 0,\ f'' : - \to +$	$f' = 0,\ f'' : + \to -$
Oblique	$0 < \lvert f' \rvert < \infty,\ - \to +$	$0 < \lvert f' \rvert < \infty,\ + \to -$
Vertical	vertical tangent, $- \to +$	vertical tangent, $+ \to -$

Do not confuse

A function can be decreasing at an *ascending* inflection point, or increasing at a *descending* one.

Quick classification exercise

Exercise Classify the point $x = 0$ for each function.

$$\text{A: } f(x) = x^3, \qquad \text{B: } g(x) = -x^3,$$
$$\text{C: } h(x) = x^3 - 2x, \quad \text{D: } p(x) = x^4,$$
$$\text{E: } q(x) = \sqrt[3]{x}.$$

For each one ask:

1. Does concavity change?
2. What type of tangent is present?
3. If it is an inflection point, is it ascending or descending?

Quick classification: solution

Function	Tangent	Classification at 0
x^3	horizontal	ascending inflection
$-x^3$	horizontal	descending inflection
$x^3 - 2x$	oblique	ascending inflection
x^4	horizontal	no inflection
$\sqrt[3]{x}$	vertical	descending inflection

Core principle

Always establish the change of concavity first. The tangent only classifies an inflection point that actually exists.

Second hour: complete function study workflow

For each function we follow the same logic:

$$\boxed{D \to \text{continuity} \to \text{zeros/sign} \to \text{limits/asymptotes} \to f' \to f'' \to \text{graph}}$$

Today the second derivative receives extra attention because we also classify every inflection point.

Study 1

We study completely

$$\boxed{f_1(x) = \exp\left(-\frac{x}{(x-2)^2}\right).}$$

Useful observation:

$$f_1(x) > 0 \qquad \forall x \in D.$$

The exponential never changes sign, so many qualitative properties will be determined by the exponent and its derivatives.

f_1: domain, intersections and sign

The denominator in the exponent cannot vanish:

$$D = \mathbb{R} \setminus \{2\}.$$

Since the exponential is strictly positive,

$$f_1(x) > 0 \quad \text{on the whole domain;}$$

there are no intersections with the x-axis. At $x = 0$,

$$f_1(0) = 1,$$

so the graph intersects the y-axis at $(0, 1)$. The function is neither even nor odd.

f_1: behaviour near $x = 2$

As $x \to 2^{\pm}$,

$$-\frac{x}{(x-2)^2} \longrightarrow -\infty,$$

therefore

$$\boxed{\lim_{x \to 2^-} f_1(x) = \lim_{x \to 2^+} f_1(x) = 0.}$$

Thus $x = 2$ is **not** a vertical asymptote.

Geometric interpretation

The graph has a removable discontinuity: it approaches the missing point $(2, 0)$. Defining $f_1(2) = 0$ would make the function continuous there.

f_1: behaviour at infinity

For $x \to \pm\infty$,

$$-\frac{x}{(x-2)^2} \sim -\frac{1}{x} \longrightarrow 0.$$

Hence

$$\boxed{\lim_{x\to\pm\infty} f_1(x) = 1.}$$

Therefore

$$\boxed{y = 1}$$

is a horizontal asymptote at both ends. More precisely, the graph approaches 1 from above as $x \to -\infty$ and from below as $x \to +\infty$.

f_1: first derivative

Let

$$g(x) = -\frac{x}{(x-2)^2}, \qquad f_1(x) = e^{g(x)}.$$

Then

$$g'(x) = \frac{x+2}{(x-2)^3},$$

and therefore

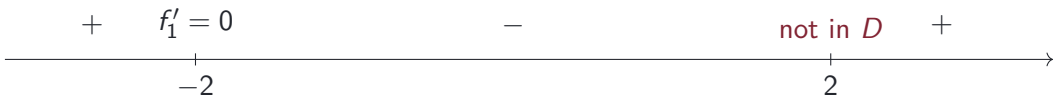
$$\boxed{f_1'(x) = e^{-x/(x-2)^2}\frac{x+2}{(x-2)^3}.}$$

Since the exponential factor is positive, the sign of f_1' is the sign of

$$\frac{x+2}{(x-2)^3}.$$

The only stationary point is $x = -2$.

f_1: monotonicity and extrema



Thus f_1:

- increases on $(-\infty, -2)$;
- decreases on $(-2, 2)$;
- increases on $(2, +\infty)$.

At $x = -2$,

$$f_1(-2) = e^{1/8},$$

so

$$\boxed{(-2, e^{1/8}) \text{ is a local and global maximum.}}$$

There is no global minimum; the infimum is 0 and is not attained.

f_1: second derivative

From

$$f_1'' = e^{g(x)}((g')^2 + g'')$$

with

$$g''(x) = -\frac{2(x+4)}{(x-2)^4},$$

we obtain

$$\boxed{f_1''(x) = e^{-x/(x-2)^2}\frac{-2x^3 + x^2 + 28x - 28}{(x-2)^6}.}$$

Both the exponential and $(x-2)^6$ are positive on the domain. Therefore the sign of f_1'' is determined entirely by

$$P(x) = -2x^3 + x^2 + 28x - 28.$$

f_1: inflection candidates

The cubic

$$P(x) = -2x^3 + x^2 + 28x - 28$$

has three real zeros, numerically

$$\boxed{\alpha \approx -3.9464, \qquad \beta \approx 1.0420, \qquad \gamma \approx 3.4044.}$$

Its sign is

x	$(-\infty, \alpha)$	(α, β)	(β, γ)	$(\gamma, +\infty)$
$P(x)$	$+$	$-$	$+$	$-$

The excluded point $x = 2$ lies inside the positive-concavity region but is not an inflection point because it is outside the domain.

f_1: concavity and inflection classification

Therefore:

- concave up on $(-\infty, \alpha)$, $(\beta, 2)$ and $(2, \gamma)$;
- concave down on (α, β) and $(\gamma, +\infty)$.

All three roots of P correspond to genuine inflection points.

x	$f_1(x)$	change of concavity	type
$\alpha \approx -3.9464$	1.1181	$+ \to -$	oblique descending
$\beta \approx 1.0420$	0.3213	$- \to +$	oblique ascending
$\gamma \approx 3.4044$	0.1780	$+ \to -$	oblique descending

Their tangent slopes are finite and non-zero.

f_1: qualitative summary

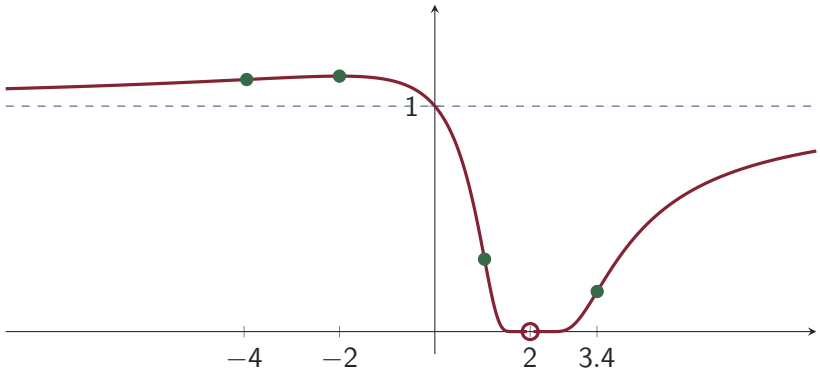
For

$$f_1(x) = \exp\left(-\frac{x}{(x-2)^2}\right):$$

- $D = \mathbb{R} \setminus \{2\}$, $f_1 > 0$;
- removable discontinuity at $(2, 0)$;
- horizontal asymptote $y = 1$ as $x \to \pm\infty$;
- global maximum $(-2, e^{1/8})$;
- three oblique inflection points: descending, ascending, descending;
- no minimum; $\inf f_1 = 0$.

The graph is now determined qualitatively.

f_1: final graph



The open point at $(2, 0)$ represents the removable discontinuity.

Study 2

Now study completely

$$\boxed{f_2(x) = \sqrt{|x-2|} - |x|.}$$

The absolute values force us to work piecewise, with breakpoints

$$\boxed{x = 0 \quad \text{and} \quad x = 2.}$$

This example is especially useful for distinguishing extrema, non-differentiability and inflection points.

f_2: piecewise representation

We have

$$f_2(x) = \begin{cases} \sqrt{2-x} + x, & x < 0, \\ \sqrt{2-x} - x, & 0 \le x < 2, \\ \sqrt{x-2} - x, & x \ge 2. \end{cases}$$

The domain is

$$\boxed{D = \mathbb{R}}$$

and the function is continuous everywhere because both $|\cdot|$ and $\sqrt{\cdot}$ are continuous on the relevant sets.

f_2: zeros and sign

Solve

$$\sqrt{|x-2|} = |x|.$$

Both sides are non-negative, so squaring is safe:

$$|x-2| = x^2.$$

The admissible solutions are

$$\boxed{x = -2, \qquad x = 1.}$$

Hence

$$f_2(x) \begin{cases} < 0, & x < -2, \\ > 0, & -2 < x < 1, \\ < 0, & x > 1. \end{cases}$$

Also

$$f_2(0) = \sqrt{2}.$$

f_2: behaviour at infinity and asymptotes

As $x \to +\infty$,

$$f_2(x) = \sqrt{x-2} - x \longrightarrow -\infty.$$

As $x \to -\infty$,

$$f_2(x) = \sqrt{2-x} + x \longrightarrow -\infty.$$

The slopes at infinity satisfy

$$\lim_{x\to+\infty} \frac{f_2(x)}{x} = -1, \qquad \lim_{x\to-\infty} \frac{f_2(x)}{x} = 1,$$

but the corresponding intercept corrections diverge:

$$f_2(x) + x = \sqrt{x-2} \to +\infty, \qquad f_2(x) - x = \sqrt{2-x} \to +\infty.$$

Therefore there are no horizontal or oblique asymptotes.

f_2: first derivative

For $x \neq 0, 2$,

$$f_2'(x) = \begin{cases} 1 - \dfrac{1}{2\sqrt{2-x}}, & x < 0, \\ -1 - \dfrac{1}{2\sqrt{2-x}}, & 0 < x < 2, \\ \dfrac{1}{2\sqrt{x-2}} - 1, & x > 2. \end{cases}$$

On $(-\infty, 0)$ the derivative is positive; on $(0, 2)$ it is negative. For $x > 2$,

$$f_2'(x) = 0 \quad \Longleftrightarrow \quad x = \frac{9}{4}.$$

f_2: what happens at $x = 0$?

At $x = 0$ the one-sided derivatives are

$$f'_{2,-}(0) = 1 - \frac{1}{2\sqrt{2}} > 0,$$

$$f'_{2,+}(0) = -1 - \frac{1}{2\sqrt{2}} < 0.$$

They are finite but different.

Conclusion

$x = 0$ is a **corner point** and a local maximum. Since

$$f_2(0) = \sqrt{2},$$

and the function tends to $-\infty$ at both ends, it is also the global maximum.

f_2: what happens at $x = 2$?

As $x \to 2^-$,

$$f_2'(x) \to -\infty,$$

whereas as $x \to 2^+$,

$$f_2'(x) \to +\infty.$$

The graph reaches

$$f_2(2) = -2.$$

Conclusion

$(2, -2)$ is a sharp local minimum (a cusp with vertical one-sided tangents).

This is a point of non-differentiability, but not automatically an inflection point.

f_2: monotonicity and extrema

The derivative sign gives

x	$(-\infty, 0)$	$(0, 2)$	$(2, 9/4)$	$(9/4, +\infty)$
f_2'	$+$	$-$	$+$	$-$

Therefore:

- global maximum at $(0, \sqrt{2})$;
- local minimum at $(2, -2)$;
- local maximum at
$$\left(\frac{9}{4}, -\frac{7}{4}\right).$$
- no global minimum, because $f_2(x) \to -\infty$ at both ends.

f_2: second derivative

For every $x \neq 0, 2$,

$$f_2''(x) = \begin{cases} -\dfrac{1}{4(2-x)^{3/2}}, & x < 0, \\ -\dfrac{1}{4(2-x)^{3/2}}, & 0 < x < 2, \\ -\dfrac{1}{4(x-2)^{3/2}}, & x > 2. \end{cases}$$

Hence

$$\boxed{f_2''(x) < 0}$$

on each differentiability interval. The graph is concave down on

$$(-\infty, 0), \qquad (0, 2), \qquad (2, +\infty).$$

f_2: are there inflection points?

There are two non-smooth candidates, $x = 0$ and $x = 2$. However, f_2'' is negative on both sides of each point:

$$\text{concave down} \quad \longrightarrow \quad \text{concave down.}$$

Therefore the concavity does **not** change.

$$\boxed{f_2 \text{ has no inflection points.}}$$

Lesson

A corner, cusp or vertical tangent is not an inflection point unless the concavity actually changes.

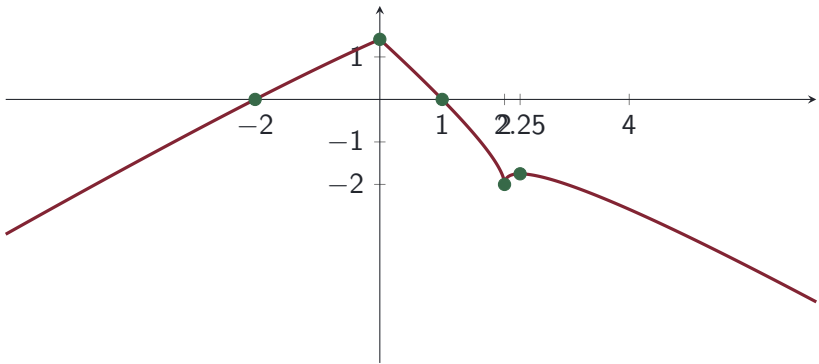
f_2: qualitative summary

For

$$f_2(x) = \sqrt{|x-2|} - |x| :$$

- $D = \mathbb{R}$, continuous everywhere;
- zeros at $x = -2$ and $x = 1$;
- no asymptotes;
- global maximum $(0, \sqrt{2})$, with a corner;
- local minimum $(2, -2)$, with a cusp;
- local maximum $(9/4, -7/4)$;
- concave down on every differentiability interval;
- no inflection points.

f_2: final graph



Notice how the two non-differentiable points have completely different local geometries.

What the two studies teach us

	f_1	f_2
Continuity issue	removable hole	none
Non-differentiability	at excluded $x = 2$	corner and cusp
Inflection points	three	none
Asymptote	horizontal $y = 1$	none
Main technical issue	cubic sign of f''	piecewise absolute values

Final message

Inflection points are about **change of concavity**. They must never be identified from $f''(x_0) = 0$ or non-differentiability alone.

Complete function-study checklist

At this point the one-variable function-study toolkit is complete:

1. domain and continuity;
2. symmetry;
3. intersections and sign;
4. limits and vertical/horizontal/oblique asymptotes;
5. f': monotonicity, stationary and non-differentiable extrema;
6. f'': concavity and inflection points;
7. inflection classification: horizontal/oblique/vertical and ascending/descending;
8. final coherent graph.

From the next lecture

Function study remains as regular practice, while the main new block moves to integration.